DECODING DATA SCIENCE

Day 2 | Teach DecoBot the Policy

RAG with LlamaIndex: split, embed, search and answer, plus a chunking lab that shows why the cut decides the answer.



TIME	YOU WILL BUILD	FILE NEEDED
About 2 hours, hands-on	A policy search engine over the DDS leave policy	DDS_Leave_Policy_v2.pdf

Three days, one assistant



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One company, one topic: leave at Decoding Data Science.

THE PROGRAMME

DAY 1 | DONE

Control the AI

System prompt and clarity. No files needed.

DAY 2 | TODAY

Give it the policy

RAG and chunking over the leave policy PDF.

DAY 3

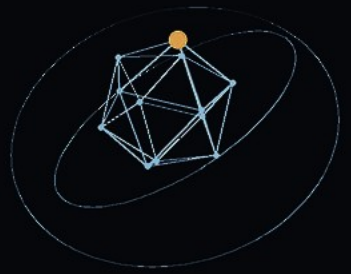
Give it data and tools

SQL, three tools and an agent that decides.

TODAY

DecoBot answers questions about the rules by searching the real policy PDF.

Today's Plan



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Six steps in about two hours. Each one is a section of the Day 2 notebook.

0:00	**Recap and setup**	Colab secret, upload the PDF, set the Gemini models
0:15	**Part A: embeddings**	How a computer finds meaning, not keywords
0:25	**Part B: first index**	Load the PDF, build an index, read the sources
0:40	**Part C: chunking lab**	Three strategies, scored on eight test questions
1:25	**Part D: policy engine**	Final prompt, scenario cards and trap questions
1:45	**Wrap-up and questions**	Checklist and tomorrow

Why We Need RAG



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A cheat sheet in the prompt worked for six facts. The real policy is much bigger.

SIZE

4 pages

The real DDS leave policy today. It could be 40 pages next year.

STRUCTURE

12 sections

Entitlements, carry-over, sick leave, parental leave, workflow and more.

THE TASK

1 question

Find the right paragraph for it, then answer using only that paragraph.

THE IDEA

Do not paste the whole policy into every question. Find the right paragraphs first, then answer from them.

RAG in Four Steps



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Retrieval augmented generation: find first, then generate.

STEP 1

Split

Cut the PDF into pieces called chunks.

STEP 2

Embed

Turn each chunk into numbers that capture its meaning.

STEP 3

Retrieve

Find the chunks closest to the question.

STEP 4

Generate

Gemini answers using only those chunks, and names the section.

THE PIPELINE

PDF > chunks > numbers > best 3 chunks > answer with the section

What Is an Embedding?



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A sentence becomes a list of numbers. Similar meaning gives similar numbers.

MEANING AS NUMBERS

An embedding is a point in space. Sentences with similar meaning sit close together.

Search means finding the closest points to the question.

A QUERY, THREE CANDIDATE SENTENCES

Annual leave: 22 working days per year	EXPECTED: CLOSEST
Medical certificate is required for sick leave	EXPECTED: FAR
The office cafeteria opens at 8 am	EXPECTED: FAR

WHY IT MATTERS

The query says "How many vacation days do I get?" and the policy says "annual leave". No shared word, same meaning. That is why RAG beats Ctrl+F.

Load the PDF, Build an Index



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Five lines of LlamaIndex give us a working first version of the search.

```
pdf = PdfReader("DDS_Leave_Policy_v2.pdf")
documents = [Document(text=page.extract_text())
             for page in pdf.pages]

index = VectorStoreIndex.from_documents(documents)
query_engine = index.as_query_engine(
    similarity_top_k=3)

response = query_engine.query(
    "How much notice for 12 working days?")
```

Lesson 1

When a RAG answer is wrong, read the retrieved chunks before you blame the model.

See the sources

show_sources() prints each retrieved chunk with its score.

Chunking: The Cut Decides



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Before search, the document is cut into chunks. How you cut decides what search can find.

TOO BIG

Mixed topics

A chunk mixes several topics, so the search gets noisy and the answer gets diluted.

TOO SMALL

Broken rules

A rule is cut in half, so the answer loses its context.

BY MEANING

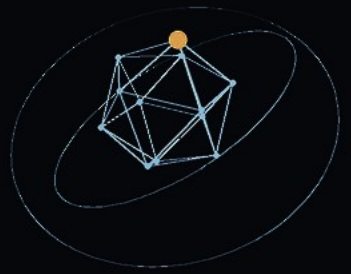
Whole ideas

Cut at the section headings, so each chunk holds one complete idea.

RULE OF THUMB

A chunk should hold one complete idea, together with its title.

Three Strategies, One PDF



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We cut the same policy three ways and compare what comes out.

STRATEGY	HOW IT CUTS	CHUNKS	AVERAGE SIZE
A Big chunks	SentenceSplitter, about 1000 tokens, no overlap	**4**	about 2,765 characters
B Small chunks	About 120 tokens with a 20 token overlap	**28**	about 416 characters
C By section	One chunk per numbered heading, title kept as metadata	**13**	about 850 characters

MEASURED ON THE REAL POLICY

These are the counts the notebook prints for the DDS leave policy. Your numbers should be close.

The Notice Table Test



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Section 3.4 sets the notice for annual leave. Is the rule still in one piece?

A | BIG

Buried

The rule sits inside a chunk of about 3,600 characters, mixed with unrelated sections.

B | SMALL

Split

The 1-2 day line ends up in a different chunk from the 3-9 and 10 or more day lines.

C | BY SECTION

Whole

The whole notice table stays together inside one section chunk of about 1,400 characters.

THE RULE IN SECTION 3.4

1-2 working days: 7 calendar days notice. 3-9 working days: 14. 10 or more: 30.

Measure, Do Not Guess



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We score each strategy on eight test questions where we know the right answer.

STEP 1

8 test questions

Each has a phrase that must appear in the retrieved chunks, for example 30 calendar days.

STEP 2

PASS or MISS

If the phrase is among the retrieved chunks, the strategy passes that question.

STEP 3

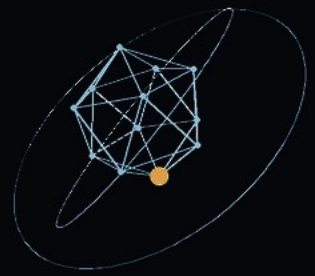
Compare

Print the hit rate for A, B and C, then change chunk size and overlap and test again.

LESSON 2

There is no best chunk size. The best strategy depends on the document. Always test it.

How Many Chunks to Retrieve?



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The setting top_k decides how many chunks the model receives.

TOO FEW | K = 1

May miss it

One chunk risks missing the answer when the rule is spread over two sections.

ABOUT RIGHT | K = 3

Enough context

A few chunks give the model what it needs with limited noise.

TOO MANY | K = 8

Noise and cost

More text costs more, slows the answer and can confuse the model.

TRY IT

Run k = 1, 2, 3 and 5 on the section strategy and compare the hit rate.

The Final Policy Engine



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Section chunks, a strict answer prompt, and one function we reuse tomorrow.

```
Answer ONLY from the policy excerpts below.
Quote numbers exactly and mention the section.
If the answer is not in the excerpts, reply:
"I could not find that in the DDS leave policy.
 Please contact People & Culture."

Keep the answer to 4 short lines.
```

1 Answer only from the policy

The prompt template forces exact numbers and a section name.

2 Say not found

If the answer is missing, the bot says so instead of inventing a rule.

3 One function

ask_policy(question) becomes a tool on Day 3.

Scenario Cards and Traps

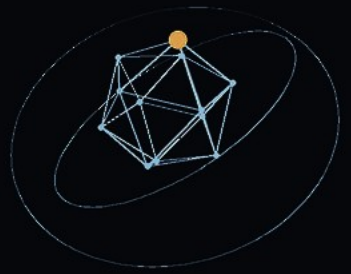


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Run each question. Check: correct, cited, and inside the policy.

QUESTION	WHAT THE POLICY SAYS
My grandmother passed away. How many days of leave do I get?	Section 8.1: 3 working days, full pay
Can I get my unused annual leave paid out in cash?	Section 4.3: not during employment
I want to go for Hajj. What is the rule?	Section 8.4: up to 30 days unpaid, once, 60 days notice
I was sick on Monday and Tuesday. What do I need to do?	Section 5.3: medical certificate for 2 or more days
TRAP: What is the dress code at DDS?	Not in the policy. The bot must say so.
TRAP: How many annual leave days do I have left?	A PDF cannot know it. That needs the database.

Wrap-Up and Tomorrow



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RAG answers what the rules are. It cannot answer what my balance is.

CHUNKING CHECKLIST

One complete idea per chunk
Section title kept with the chunk
Measured with test questions
top_k big enough, not noisy
Says "I do not know" when it should

TOMORROW: DAY 3

Give DecoBot data and tools

SQL on the DDS leave database, three tools, and an agent that decides which one to use. It can then answer "what is MY balance?"